Your Name:

ID #:

Worksheet: Limits to Infinity-Part 2

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Show all the work to find the following limits.

   (a) $\lim_{x\to-\infty}(e^{3x} - e^{2x})$

   (b) $\lim_{x\to\infty} \sin(\frac{x^2}{2x^2+x})$

   (c) $\lim_{x\to\infty}(\sin x)e^{-x}$

   (d) $\lim_{x\to\infty} x \sin x$

   (e) $\lim_{x\to\infty} x \ln \frac{1}{x}$

   (f) $\lim_{x\to-\infty} xe^{\frac{1}{x}}$

2. Show all the work to find the following limits:

   (a) $\lim_{x \to \infty} x^{\frac{1}{x}}$

   (b) $\lim_{x \to \infty} (\frac{1}{x})^{\frac{1}{x}}$

   (c) $\lim_{x \to \infty} (\frac{x}{x+1})^{x}$